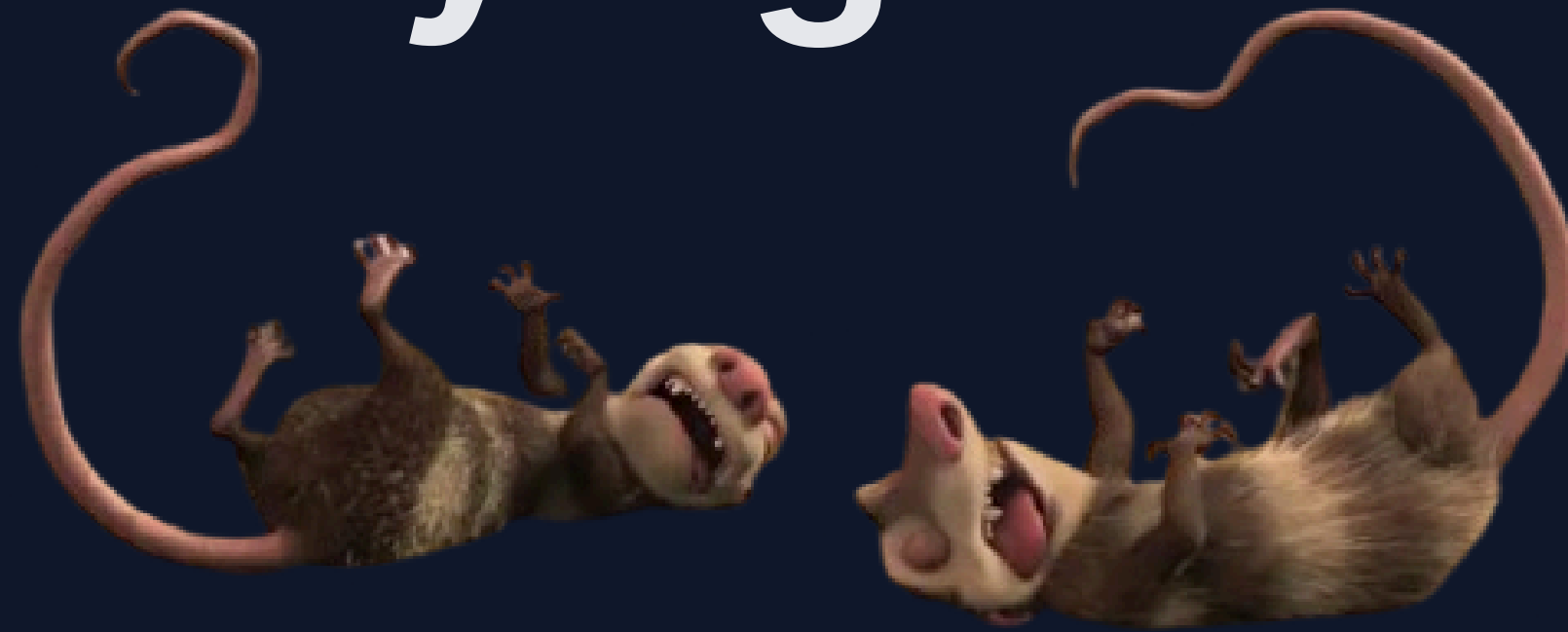


Playing Dead?



Anti-Analysis techniques of Malware

Monika Sahu

Cybersecurity Consultant

null Hyderabad

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About me

Monika Sahu

Cyber Security Consultant

~ 1.5 yrs experience

Open to learn and network, active in cybersec communities.

Content

- What is a malware?
- Analysis!
 - Aims, Techniques, Conclusion
- Type of Analysis
- Analysis,,,,,, are you sure?
- About the Topic
- Why anti-analysis?
- Types of Anti-Analysis Tricks:
- Static/Code Analysis Level
- Device - level
- Behavioral Tricks
- Approach by the analyst

What is a malware?

- What it is?
- Why does it exist?
- Types
- The impact
- What are we doing for this?
- Then what's the problem here?

Analysis!

- Aim
- Techniques
- Conclusion

Type of Analysis

- Static
 - What does the code say?
- Behavioral
 - What does it do, if I run this
- Memory
 - What happens in memory in runtime
- Post-execution artifacts
 - What impact, artifacts it left

Analysis,,,,,, are you sure?

- Malware being dead?
 - Stop executing as expected
 - Stop Executing at all!!!

Time ...	Process Name	PID	Operation	Path	Result	Detail
10:57:...	a397.exe	4464	Process Start		SUCCESS	Parent PID: 3832, ...
10:57:...	a397.exe	4464	Thread Create		SUCCESS	Thread ID: 6560
10:57:...	a397.exe	4464	Load Image	C:\Users\verse\Desktop\samples\va397.exe	SUCCESS	Image Base: 0x100...
10:57:...	a397.exe	4464	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7ffc...
10:57:...	a397.exe	4464	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x775...
10:57:...	a397.exe	4464	CreateFile	C:\Windows\Prefetch\A397.EXE-C83EBBDE.pf	NAME NOT FOUND	Desired Access: G...
10:57:...	a397.exe	4464	Thread Exit		SUCCESS	Thread ID: 6560, ...
10:57:...	a397.exe	4464	Process Exit		SUCCESS	Exit Status: 0, User...
10:57:...	a397.exe	4464	RegOpenKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-18724396...	SUCCESS	Desired Access: All...
10:57:...	a397.exe	4464	RegQueryValue	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-18724396...	NAME NOT FOUND	Length: 40
10:57:...	a397.exe	4464	RegCloseKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-18724396...	SUCCESS	

????

About the Topic

What's the topic are we talking about?

Why anti-analysis?

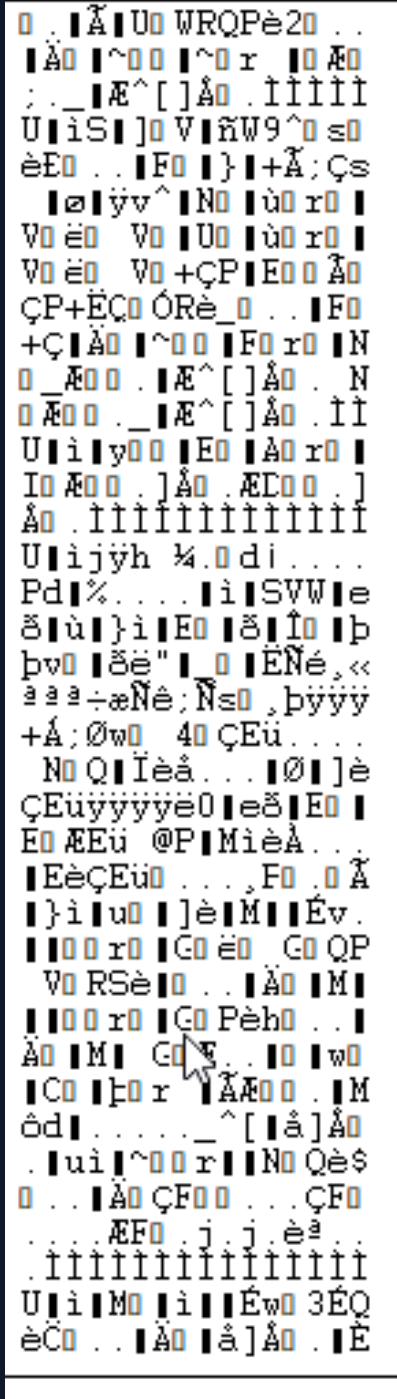
- To Evade Detection by Security Products
- To Delay Reverse Engineering
- To Enable Targeted Attacks (Selective Execution)
- To Increase Campaign Longevity

Types of Anti-Analysis Tricks:

- Static/Code Analysis Level
 - Static - level
- Device - level
 - To sense the security beams
- Behavioral tricks
 - To evade detection

Static/Code Analysis Level

- Obfuscation, Packing, Compression, Encryption
- Timestamp
- Explicit Import/Export
- Code Fragmentation
- Rabbit - Hole!



file > first 32 bytes (hex)	4D 5A 95 A4 39 E5 15 E1 BD 15 F3 A5 3F 00 89 22
file > first 32 bytes (text)	MZ....9.....?....")".."....."..s...."..
file > info	size: 188416 bytes, entropy: 7.279
file > type	executable, 32-bit, GUI
file > version	n/a

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\verse\desktop\samples\397.exe (read-only)

file settings about

c:\users\verse\desktop\samples\397.exe

- indicators (certificate > signature-info)
- footprints (type > sha256)
- virustotal (offline)
- dos-header (size > 64 bytes)
- dos-stub (size > 160 bytes)
- rich-header (checksum > invalid)
- file-header (executable > 32-bit)
- optional-header (subsystem > Native)
- directories (count > 6)
- sections (count > 6)
- libraries (count > 2)
- imports (flag > 9)
- exports (n/a)
- thread-local-storage (n/a)
- .NET (n/a)
- resources (signature > version)
- strings (count > 500)
- debug (n/a)
- manifest (n/a)
- version (FileDescription > change me)
- certificate (signature-info > issue)

library (2)	flag (0)	type	imports (40)	description
ntoskrnl.exe	-	Implicit	37	NT Kernel and System
HAL.dll	-	Implicit	3	Hardware Abstraction Layer

sha256 > A3971683F5AFA60B740AF061F36FC0D2D550073A9A2E58BD03D9AAFF69CA9937 | cpu > 32-bit | file > type > executable | subsystem > Nati

Device - level

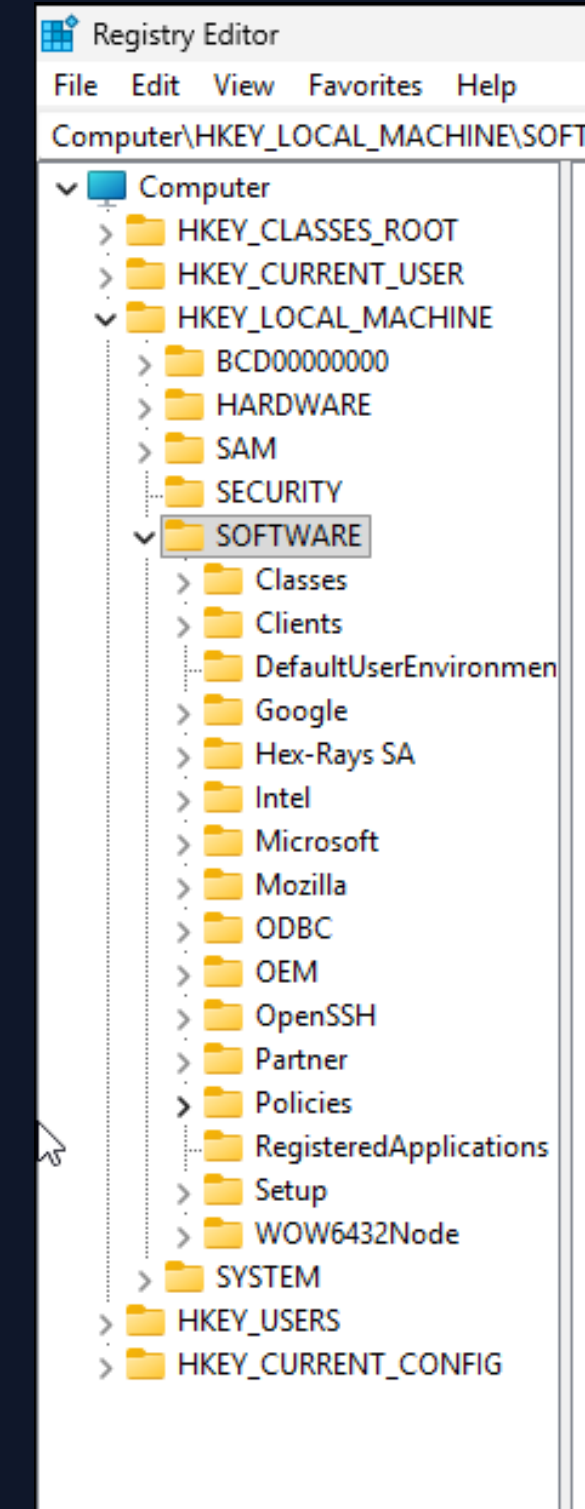
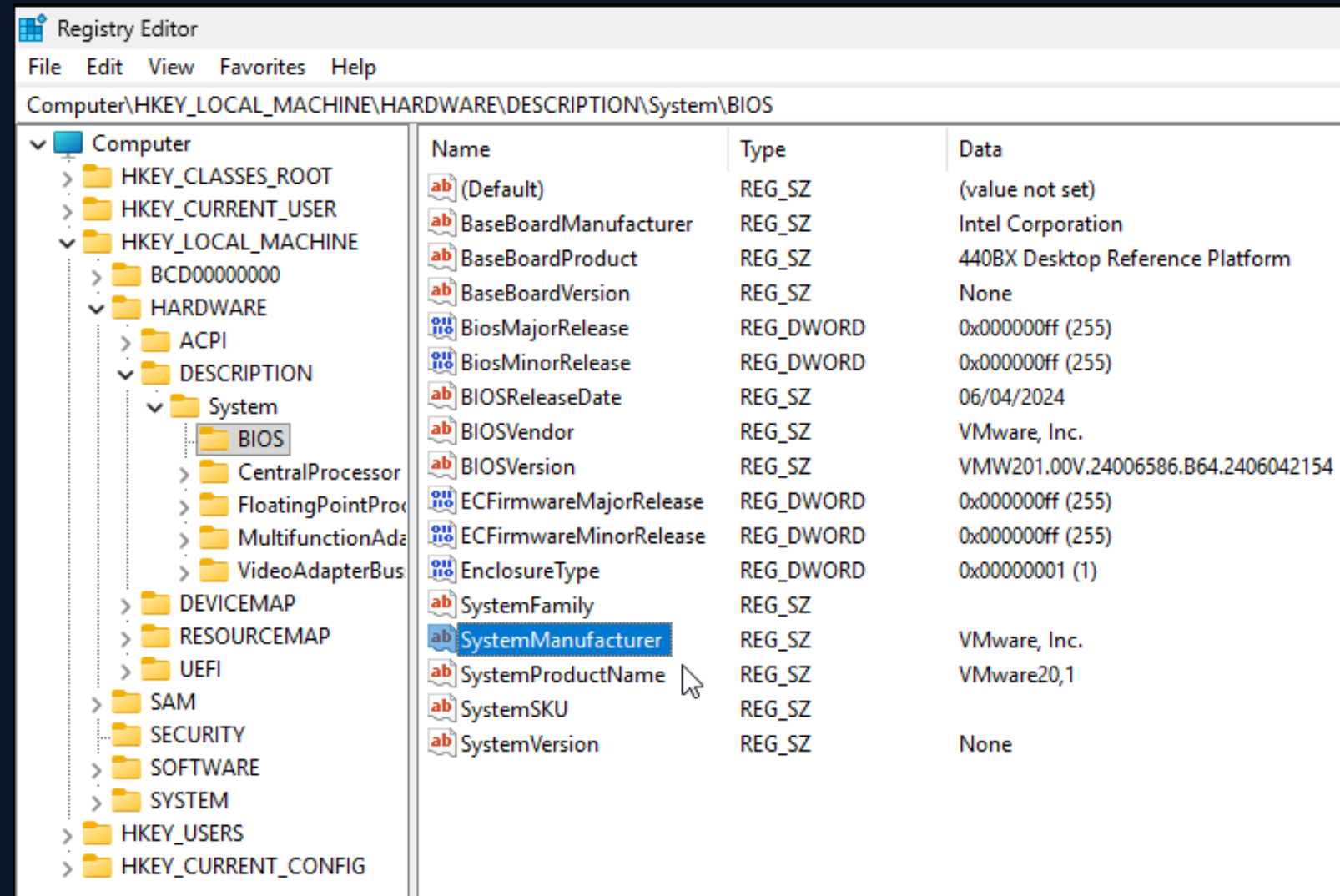
- Checking for
 - Environment of execution
 - Names/Username
 - Resources
 - Network connectivity
 - Basic Software
 - Geo-Location/IP
 - Language Settings
 - User interaction
 - Oops!! A Debugger??

Device - level

- Environment of execution
 - By default MAC Prefix, Registry exploration, Software present, Resources

```
C:\Users\verse>getmac
```

Physical Address	Transport Name
00-0C-29-A8-89-C9	\Device\Tcpip_{633AD573-E2C9-45CA-A32C-193A4C1903B4}



- Names/Username

```

f (GetUserNameA(username, &size))
{
    const char* blacklist[] = {
        "admin", "user", "test", "sandbox", "analyst",
        "vmuser", "malware", "reverse", "lab"
    };

    for (auto name : blacklist)
    {
        if (_stricmp(username, name) == 0)
            return true;
    }
}

```

- NetUserEnum
- Get-LocalUser | Select Name - fileless malwares

Symbol	Type	String
0000A	C	GET_GRAPH
0000D	C	GetLocalTime
00011	C	GetComputerNameA
00014	C	GetSystemDirectoryW
0000F	C	GetConsoleMode
00018	C	GetEnvironmentVariableW
0000D	C	GetStdHandle
00010	C	GetCommandLineA
00014	C	GetCurrentProcessId
00013	C	GetFileAttributesA
0000D	C	GetTickCount
00011	C	GetModuleHandleA
00011	C	GetModuleHandleW
00014	C	GetSystemDirectoryA
00011	C	GetTempFileNameA
0000D	C	GetLastError
0000E	C	GetVersionExA
00011	C	GetClipboardData
0000E	C	GetWindowRect
0000A	C	GetParent
0000F	C	GetWindowTextA
0000E	C	GetClientRect
00014	C	GetForegroundWindow
0000D	C	GetCursorPos
00006	C	GetDC
00011	C	GetKeyboardState

Network connectivity



```
call    InternetGetConnectedState
test    eax, eax
jz      exit_or_sleep
```

```
BOOL InternetGetConnectedState(
    LPDWORD lpdwFlags,
    DWORD   dwReserved
);
```

EAX = 1 → Internet is available
EAX = 0 → No Internet connection

```
test eax, eax
```

This is a bitwise AND of eax with itself
It does **not** change EAX
It only sets CPU flags:

ZF (Zero Flag) = 1 if EAX == 0

ZF = 0 if EAX != 0

```
jz exit_or_sleep
```

jz = jump if zero flag is set

So the jump is taken only when no internet is detected

- Geo-Location/IP, Language Settings
 - Generally used by specific threat groups, APT.

<u>SetFilePointer</u>	-	implicit	-
<u>GetTimeZoneInformation</u>	-	implicit	-
<u>GetDateFormatW</u>	-	implicit	-
<u>GetTimeFormatW</u>	-	implicit	-
<u>LCMapStringW</u>		implicit	

- User-Interaction

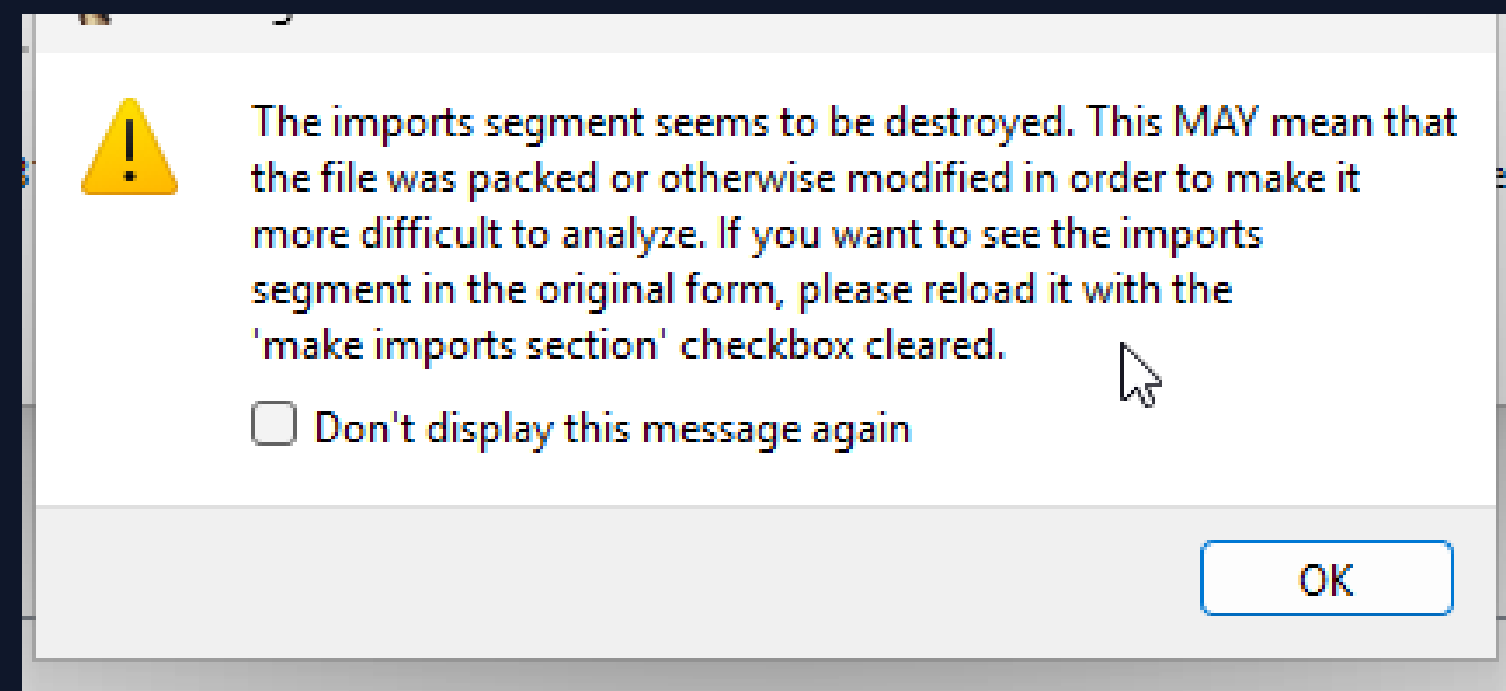
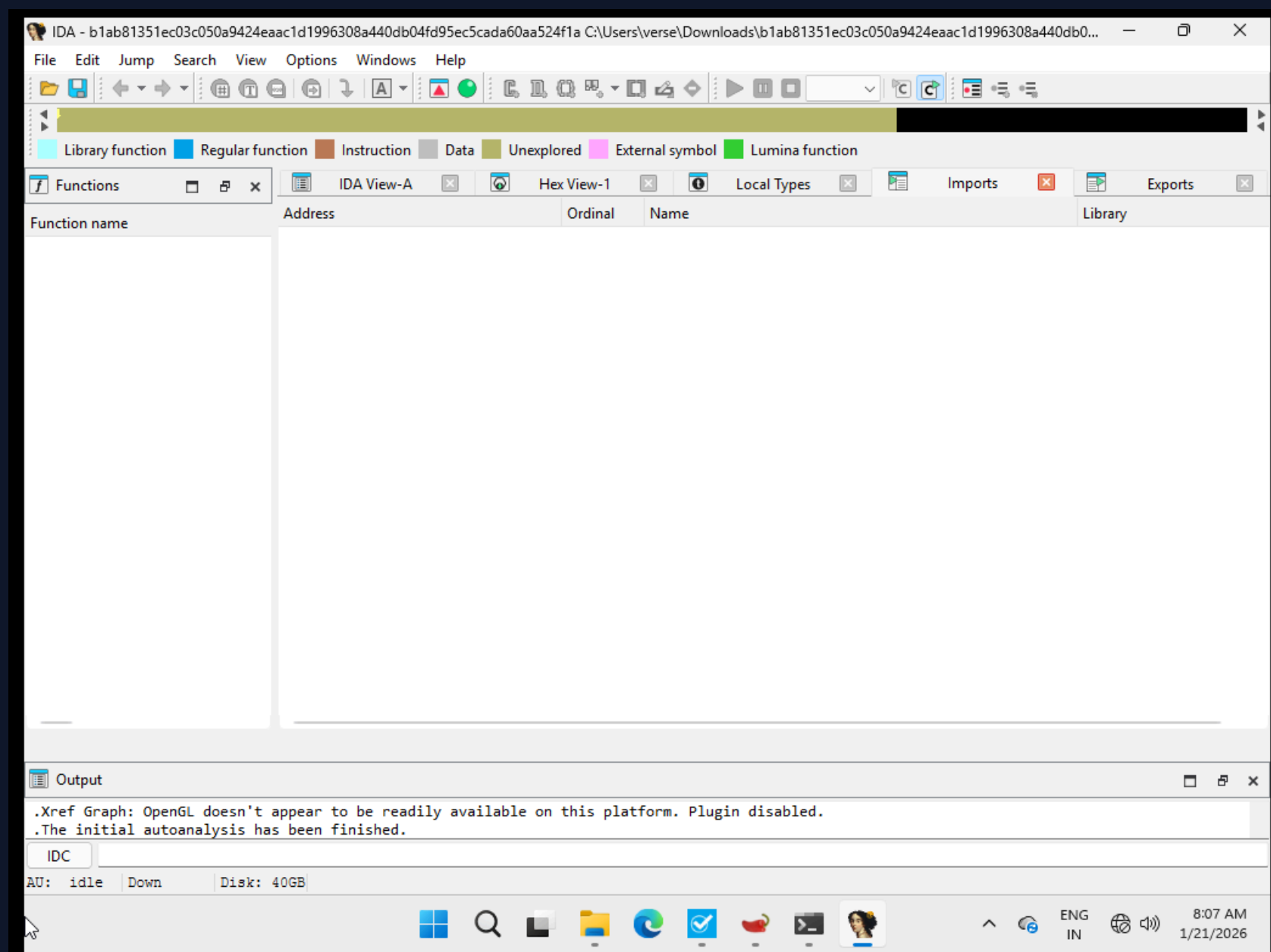
0048D57C	0048D57C	0048D57C	0048D57C
0048D580	0048D580	0048D580	mouse_event
0048D584	0048D584	0048D584	ExitWindowsEx
0048D588	0048D588	0048D588	SetActiveWindow
0048D58C	0048D58C	0048D58C	FindWindowExW
0048D590	0048D590	0048D590	EnumThreadWindows
0048D594	0048D594	0048D594	SetMenuDefaultItem
0048D598	0048D598	0048D598	InsertMenuItemW
0048D59C	0048D59C	0048D59C	IsMenu
0048D5A0	0048D5A0	0048D5A0	TrackPopupMenuEx
0048D5A4	0048D5A4	0048D5A4	GetCursorPos

- Oops!! A Debugger??

- Common Debuggers
- Check Parent
- Small sleep before main stage - GetTickCount, QueryPerformanceCounter, Sleep
- Detects if a debugger is already attached before main logic starts.
 - mainCRTStartup, WinMain, more early → TLS Callback (NtQueryInformationProcess)
- RDTSC, cpuid
- What if your system itself is slow?

0048D230	OutputDebugStringW	KERNEL32
0048D31C	IsDebuggerPresent	KERNEL32

SetCurrentDirectoryW	-	implicit
IsDebuggerPresent	-	implicit
GetCurrentDirectoryW	-	implicit



Behavioral Tricks

- Stop execution
- For acting innocent:
 - Code Redirection
 - Change in execution paths, renaming
 - Each time new surprise - Polymorphism!
 - Exception Handling
 - Reflective - DLLs/Process injection
 - Self deletion/ termination

Dynamic Analysis Evasion

Manual Dynamic Analysis Evasion (Anti-Debugger)

Automated Dynamic Analysis Evasion (Sandbox Evasion)

Direct Detection

- Parent Check
- Reading PEB
- Detecting Breakpoints
- Mining NTQueryObject
- Reading System Artifacts

Deductive Detection

- Traps
- Timing-based detection

Debugger Escaping

- Lockout Attacks
- Debugger-Specific
- Control Flow Manipulation
- Fileless (AVT) attacks

Detection-Dependant

- Targeted
- Fingerprinting
- Reverse Turing Test

Detection-Independent

- Stalling
- Trigger-based
- Fileless (AVT) attacks

1. Manual Dynamic Analysis Evasion

1.1 Direct Detection

- 1.1.1 Parent Check
- 1.1.2 Reading PEB
- 1.1.3 Detection Breakpoints
- 1.1.4 Mining NTQueryObject
- 1.1.5 Reading System Artifacts

Enumerates running processes

Finds parent PID

Compares parent name to debugger list

Reads kernel-provided flag directly

No API required

Malware scans its own code section for 0xCC (INT3), inserted by debuggers.

- Walks .text section
- Detects software breakpoints

if (*ptr == 0xCC) ExitProcess(0);

Checks internal process flags exposed by the kernel.

1. Manual Dynamic Analysis Evasion

1.2 Deductive Detection

1.2.1 Traps

1.2.2 Time-based Detection

Malware intentionally triggers exceptions and checks whether they are handled correctly.

Debuggers often intercept or alter exception flow.

```
AddVectoredExceptionHandler  
SetUnhandledExceptionFilter
```

```
start = GetTickCount();  
SensitiveFunction();  
if (GetTickCount() - start > 1000)  
    ExitProcess(0);
```

1. Manual Dynamic Analysis Evasion

1.3 Debugger Escaping

1.3.1 Lockout attacks

1.3.2 Debugger Specific

1.3.3 Control Flow Manipulation

1.3.4 Fileless AVT Attacks

Infinite loops on breakpoints

Thread starvation

Debugger deadlocks

CPU-burning loops

```
while (IsDebuggerPresent()) {  
    Sleep(1000000);  
}
```

Malware targets known bugs or behaviors of specific debuggers.

```
xor eax, eax  
jz fake_path  
jmp real_path
```

Exception-driven execution

Opaque predicates

Computed jumps

Self-modifying code

2. Automated Dynamic Analysis Evasion

2.1 Detection Dependent

2.1.1 Targetted

2.1.2 Fingerprinting

2.1.3 Control Flow Manipulation

2.1.4 Reverse Turing Test

API Calls should not be revealed.
For this? Hook Evasion

Raw syscalls invokes

2. Automated Dynamic Analysis Evasion

2.2 Detection Independent

2.2.1 Stalling

2.2.2 Trigger Based

2.2.4 Fileless AVT Attacks

Sleep

NtDelayExecution

WaitForSingleObject

Sleep(600000); // 10 minutes

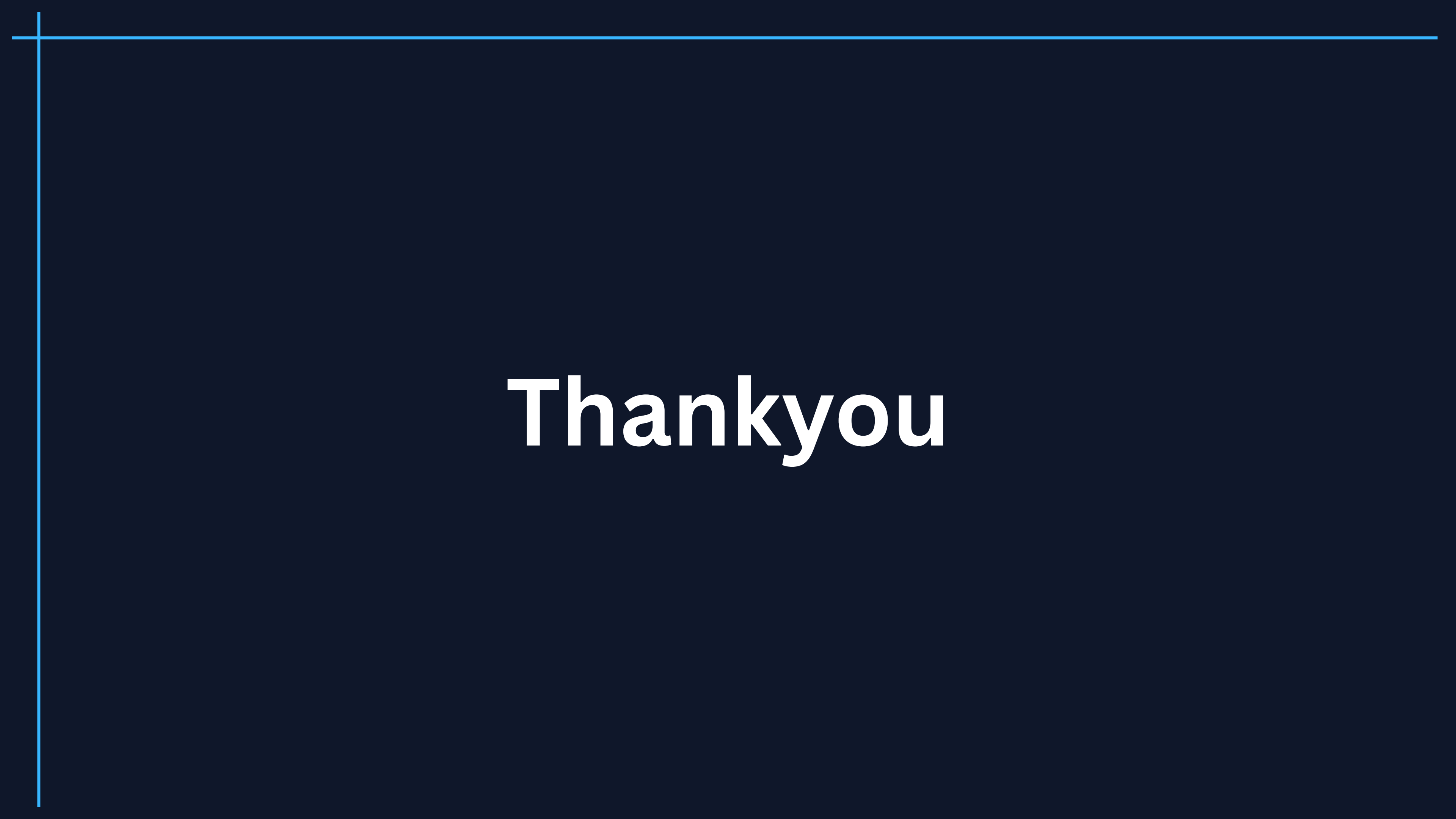
GetSystemTime

GetFileAttributesW

InternetCheckConnectionW

As an analyst

- Naming convention
- Mimic the environment config.
- Make dummy logs
- FPs > FNs
- Strong stable tools
- Hardware breakpoints



Thankyou



Monika Sahu

Security Researcher | Malware Analysis
| EDR

